

ED ML LoS vs Competitor Gate Engine — One-Pager

Purpose

Give decision-makers a crisp comparison between the **ML-only LoS→occupancy** approach and the **MLP + deterministic gate engine** approach, plus a recommended hybrid.

Snapshot

Your project — ML LoS → Occupancy - Predict **patient-level LoS** (CatBoost) from ED features and **aggregate** to hospital-wide bed forecasts (24–72h).

- Key insight: **error symmetry** in LoS often improves occupancy more than simply lowering MAE.
- Future arrivals filled with **averages**; evaluates hospital-wide occupancy (not ward-specific).

Competitor — MLP + Gate Engine - ML outputs **admission risk + LoS buckets**; a **deterministic gate engine** turns this into **policy-bound actions** (priority, next_action, TTL, reason).

- ML is **advice-only**; gates enforce **safety & explainability**; includes ops panel/nudges, label spec for an actionable target.

Head-to-Head

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Primary output	Beds forecast (H=24/48/72h)	Actionable queue rules (priorities, timers, reasons)
Unit of decision	Hospital level (aggregated)	Patient/task level (operational)
Future arrivals	Constant average fill (calendar effects missing)	Not modeled yet (focus on gates)
Granularity	Hospital-wide; no ward/acuity isolation	Per-patient policy; no occupancy total
Governance	No rule layer; no fairness audit	Read-only ML; deterministic gates + guardrails
Calibration	Not reported; post-hoc symmetrization sims	Calibrator in place; shift-aware fallbacks
Data hygiene	Standard ED features; long-stay truncation	Explicit no-leakage (features $\leq t_0$)

Dimension	ML LoS → Occupancy	MLP + Gate Engine
Ops readiness	Good for capacity planning; lacks ops hooks	Good for day-to-day flow; lacks occupancy

Strengths & Gaps

ML LoS → Occupancy

- + Converts patient predictions into bed needs; quantifies what error properties matter.
- Average-fill arrivals, no ward/acuity split, tail truncation; no rule/governance layer.

MLP + Gate Engine

- + Clear safety contract; policy knobs; explainability; ops panel/nudges.
- No hospital-wide occupancy; some label/calibration maturity still needed.

When to Use Which

- **Bed planning 24–72h out:** choose **ML LoS→occupancy** (add calendar-aware arrivals).
- **Running the ED today (escalations, timers, nudges):** choose **Gate Engine** (ML as calibrated hints).

Recommended Hybrid (Best of Both)

- 1) **Keep the occupancy aggregator;** replace average-fill with **calendar-aware arrivals** (day-of-week/season/case-mix).
- 2) **Adopt the gate engine** as a policy layer: define a clear positive label/time window (Gate_Pos v2), costs, exclusions; keep ML **read-only**.
- 3) **Calibrate & audit:** reliability curves, Brier/ACE, and **fairness slices** (sex, age, acuity, arrival hour, language, insurance).
- 4) **Wire together:** feed **LoS quantiles + admit risk** into gates; publish **ward/acuity/isolation** occupancy for staffing.

Pilot & Metrics

Shadow-mode → capped pilot with governance sign-off. Track: - **Beds MAE@H** (H=24/48/72) per ward + P95, **overflow/shortfall days**, signed bias.

- **Nudge utility:** TP – λ ·FP – μ ·FN; clinician “Not helpful” rate; override counts.
- **Flow:** boarding time, diversion hours, elective cancellations avoided.
- **Fairness:** error/calibration parity across subgroups.

Decision Guidance (TL;DR)

- Need **explainable, policy-controlled actions now** → start with **Gate Engine**.
 - Need **reliable capacity forecasts** → lean on **ML LoS→occupancy** with better arrivals.
 - Need **both** → implement the **hybrid** above; pilot in shadow-mode, then A/B with decision-centric metrics.
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Extended Analysis & Strategy

Engineering Maturity — Verdict

- **More production-ready for per-patient ops:** the gate-engine stack (read-only ML → deterministic rules, invariants, leakage checks, calibration fallback, ops panel, staged rollout).
- **More scientifically novel for capacity:** the LoS→occupancy translator and the finding that **symmetric LoS errors** reduce occupancy bias more than simply shrinking MAE.

Credibility & Conference Framing

- They're only a threat if you present the *same* problem. Keep the split clear: **who to act on now (gates)** vs. **how many/where in 24–72h (capacity)**.
- Be candid about two gaps (average-fill arrivals; artificial timeline) and state fixes. Match their safety posture: **ML advice-only for ops; policy lives in rules**.

If They Add Capacity — Why It's Non-Trivial

- They need a **calendar-aware arrivals model** (DoW/season/holiday + case-mix) to replace averages; otherwise time-dependent error remains high.
- **Aggregation quality ≠ point accuracy:** they must optimize for **symmetric signed errors** and low occupancy bias, not just LoS MAE.
- **Calibration under shift:** turning bucketed probabilities into reliable occupancy distributions with uncertainty bands.
- **Granularity:** ward/acuity/isolation forecasts and translation into **staffable rosters** (skill-mix, blocked beds).
- **Evaluation:** decision-centric metrics (Beds MAE@H, P95 error, overflow/shortfall days, signed bias) and external validation with real timestamps.
- **Fairness & privacy:** subgroup performance, guardrails on sensitive features, synthetic-data privacy tests.

Defensibility Moves (Next 4 Weeks)

1) **Ship arrivals:** calendar-aware arrivals; show delta vs. average-fill on 24/48/72h MAE and signed bias. 2) **Train for aggregation:** quantile/pinball loss; add a symmetry regularizer; report occupancy MAE@H, P95, signed bias, overflow days; add isotonic/quantile calibration. 3) **Go granular:** ward/acuity/isolation forecasts; map to staffing plans and skill-mix; include blocked-bed logic. 4) **Governance parity:** advice-only ML for ops; deterministic gate layer for nudges; reliability curves, fairness slices, drift monitoring, override

logging. 5) **Rare cohorts**: cap synthetic:real ($\leq 1:1$), utility-filter synthetic, run privacy audits; use DP-CTGAN for any external sharing.

Conference Talk Track & Q&A (Scriptable)

- **Open with problem split**: Capacity vs. Ops; show two pipelines and decisions.
- **Claim unique contributions**: LoS→occupancy translator; **error-symmetry** impact on occupancy.
- **Show fixes**: arrivals model replacing averages; ward-level outputs; calibration + fairness dashboards.
- **Safety alignment**: read-only ML for ops; deterministic policy gates.
- **Q&A snippets**:
 - “*Why no gate engine?*” We isolate capacity; for ops we use read-only ML and rules—same safety contract.
 - “*Seasonality/arrivals?*” We replaced averages with calendar-aware arrivals; that’s the main driver of time-dependent accuracy.
 - “*Calibration/fairness?*” Reliability curves + subgroup slices; same calibrator-with-fallback pattern for ops surfaces.
 - “*Could they do capacity?*” Yes, with arrivals modeling, aggregation-aware training, calibration, and ward-level translation—this is exactly what we’re shipping now.

Bottom Line

Use their **gate engine** for governed, explainable per-patient actions; keep and harden your **capacity module** for reliable 24–72h staffing and beds. Present them as **complementary** and lead with concrete fixes and decision-centric metrics.

Competitive Stance & Messaging (Concise Addendum)

- **Take them seriously—don’t dismiss**. Their read-only ML → deterministic **gate engine** is credible ops engineering. Your lane is **capacity science** (LoS→ward-level occupancy + the **error-symmetry** insight).
- **Credibility risk if dismissed**: slick workflow demos can make you look defensive. Neutral framing = *complementary layers* that interoperate.

Do next (30 days)

- **Calendar-aware arrivals** (DoW/season/holiday + case-mix); report ΔBeds **MAE@24/48/72h** and **signed bias**.
- **Ward/acuity/isolation forecasts** with uncertainty bands; map to **staffable rosters** (skill mix, blocked beds).
- **Governance parity**: ML **advice-only**, deterministic rules for policy; leakage checks, reliability curves, and calibrator with fallback.
- **Hybrid offer**: feed **LoS quantiles + ward forecasts** into any gate engine for better escalations and reduced boarding.

Messaging cues

- **Use**: complementary, interoperate, decision layers, calibrated, governed.

- **Avoid:** better/worse, demo-only, “they can’t.”

If facts change

- **They show measured outcomes (boarding/LoS/safety):** integrate and **co-report** joint metrics.
- **They remain demo-level:** keep the complementary framing; lead with **decision-centric metrics** and your ward-level outputs.

Bottom line

Treat them as **serious, complementary competition**. Protect credibility by owning **capacity forecasting** and matching **governance** for ops nudges—so patients and staff see concrete benefits.

Gate_Pos (Competitor) — Concise Assessment

Strengths - Solid leakage hygiene (features $\leq t$) and timestamped snapshots; clear horizons. - Multi-task trunk w/ appropriate heads (complexity, resource, 4h deterioration, time-to-readiness, workup completion). - Correct ordinal modeling for T2/T4; basic calibration plan; group reporting. - Read-only ML → deterministic gate integration; safety preserved via `priority = max(gate_floor, f(...))`.

Key Risks - **Action-defined labels (T2):** resource use reflects crowding/practice patterns → bias. - **T1 scope creep:** mixing future 2h flags with present calculators blurs construct; correlates with T3. - **Censoring:** dropping early exits biases T3/T4; needs time-to-event handling. - **Plan extraction fragility:** readiness/completion depend on noisy order/result/consult streams. - **Calibration under shift:** ordinal buckets/iso cutpoints can break across day/night, weekend. - **Multi-task interference:** one head can dominate; gradient conflict. - **Priority thrash:** 15–30 min re-scores can cause queue flapping without hysteresis. - **Fairness coverage:** coarse groups miss intersections (sex×age×arrival hour×acuity, language/insurance proxies). - **Eval not decision-centric:** AUC alone won’t convince ops; need recall@K/utility. - **Interpretability drift:** per-case saliency unstable; need reason strings aligned to calculators.

How it pairs with capacity - Excellent for **who now**; combine with our **calendar-aware arrivals** + LoS→ward occupancy for **how many/where next (24–72h)**.

Hardening Checklist (for Q&A & integration)

- **Deconfound T2:** control for shift/crowding; add negative controls; report by occupancy slices.
- **Purify T1:** base on present calculators/physiology only; move future flags to T3.
- **Time-to-event:** discrete-time hazard/survival heads for T3; ordinal-survival or quantile for T4; keep censored data.
- **Plan extractor:** versioned care-plans, explicit waive events, 30–60 min stability guard; log denominator changes.
- **Calibration:** monotone-constrained ordinal calibration; maintain day/night & weekday/weekend calibrators; publish ranks + uncertainty if collapse.
- **Multi-task training:** PCGrad/GradNorm or alternating head updates; monitor per-head gradient norms.
- **Queue stability:** TTL/hysteresis & rate-limit re-ordering; budgeted top-K per zone.

- **Fairness:** intersecting slices (sex×age×arrival hour×acuity; language/insurance where allowed); group-wise calibration; track FNR gaps for T3.
- **Decision metrics:** recall@K and Net Benefit/utility ($TP - \lambda \cdot FP - \mu \cdot FN$) for T3; median & P95 abs error for T4; operational KPIs (boarding, diversion).
- **Interpretability:** deterministic reason strings from heads + calculators; stable, bin-level attributions (e.g., TreeSHAP on monotone bins).
- **Missingness & drift:** explicit missing indicators, last-value age; drift monitoring on inputs/labels; override logging.
- **Uncertainty:** conformal intervals for T3/T4; display bands in ops UI.
- **Policy sensitivity:** simulate gate TTL/threshold variations; show robustness of rankings.

Use this checklist to position a **safe integration**: their Gate_Pos for governed per-patient actions; our ward-level forecasts to set expectations and staffing 24–72h out.